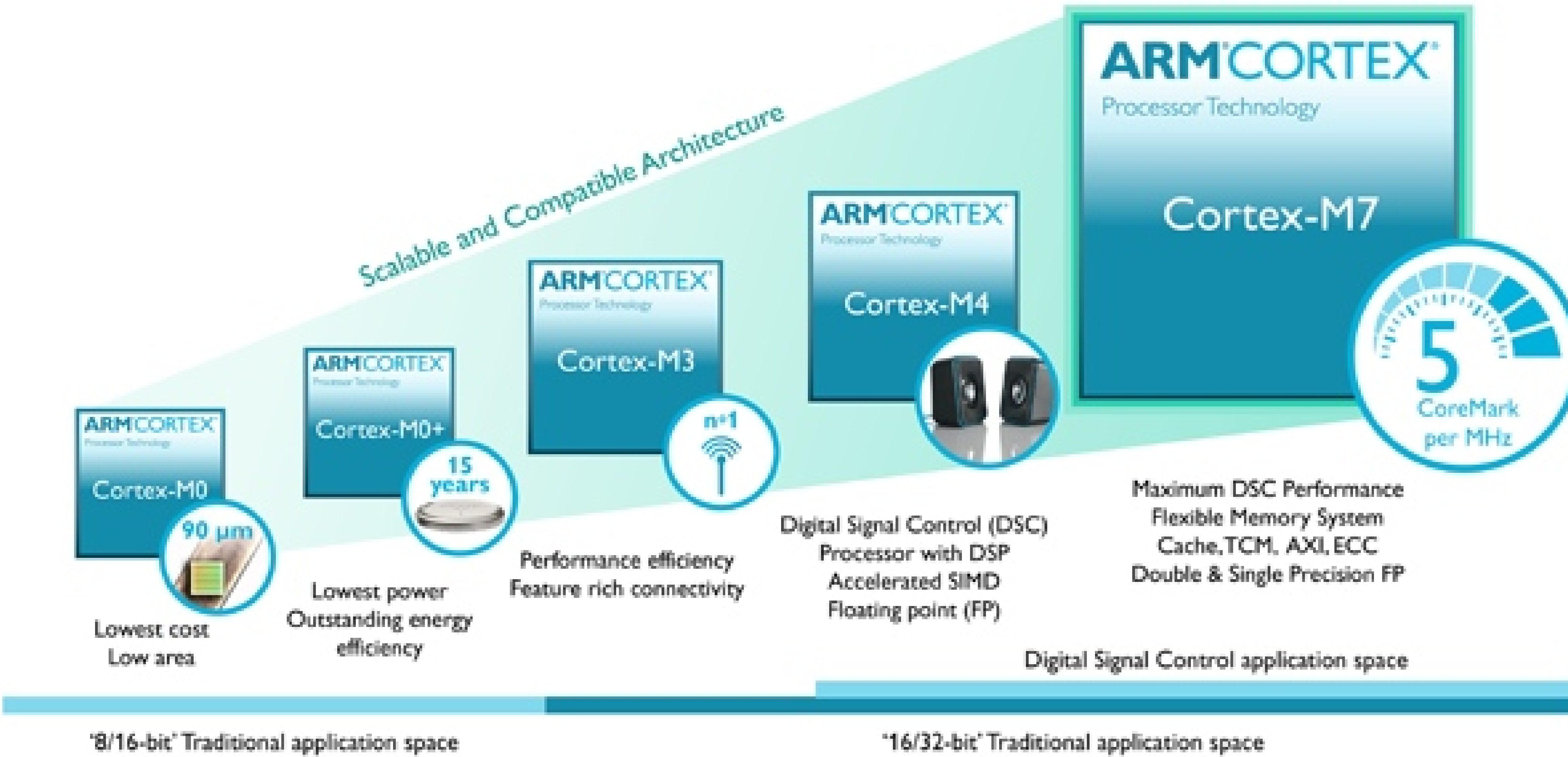
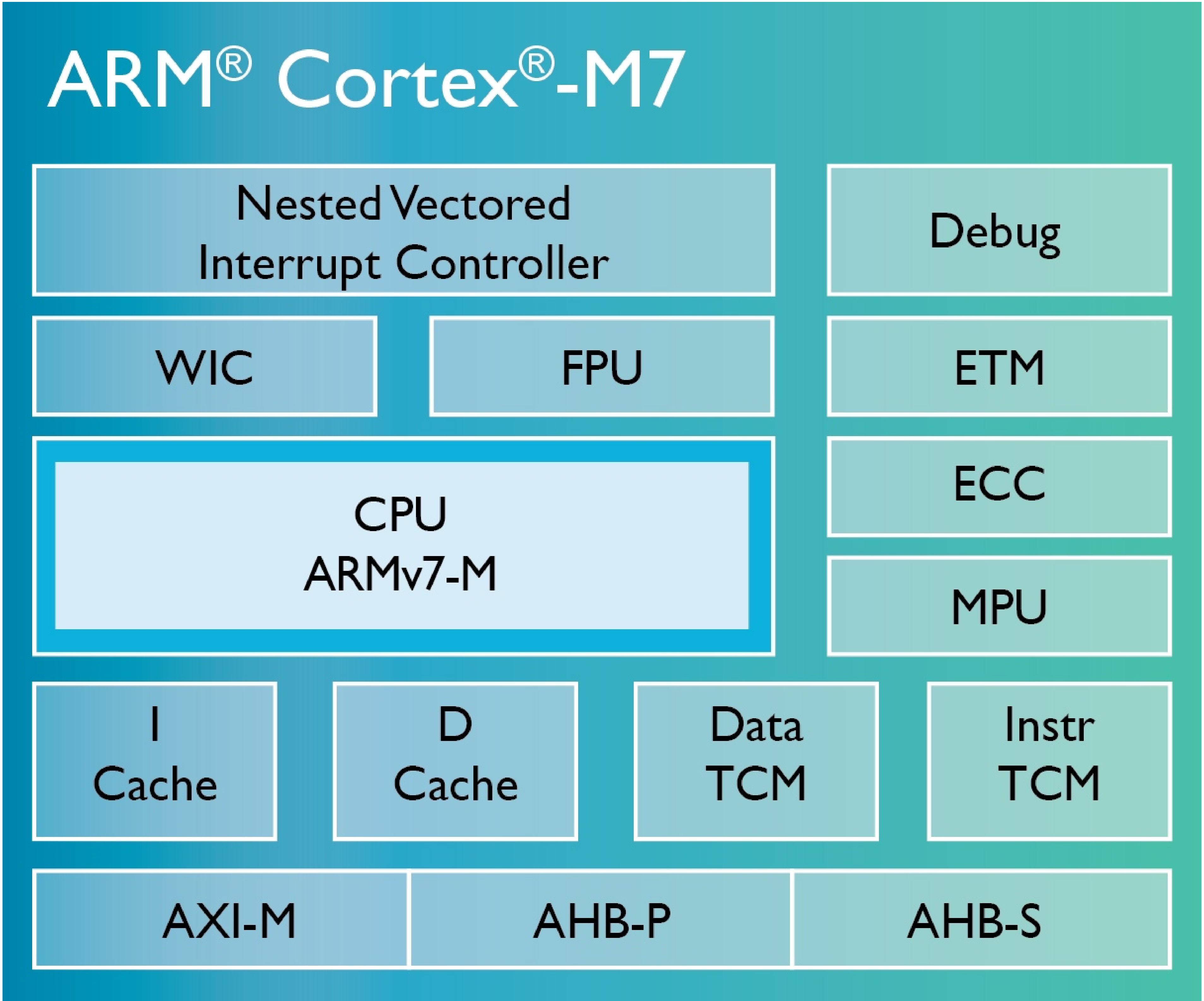
Lecture Overview

- Introduction, Microcontroller basics
- 32-bit Microcontroller technology
- ARM Cortex M Architecture, (CM0 CM3 CM4 CM7)
- Cortex M Microcontroller Instruction Set
- Exception Handling, Interrupts
- Real-time Operating Systems (RTOS)
- Memory Management, protected memory, MPU
- Implementing Calculations, Data Formats, FPU
- I/O, Device drivers
- Application Design
- Case Studies, Advanced Design Patterns

https://skript.eit.h-da.de/ams/



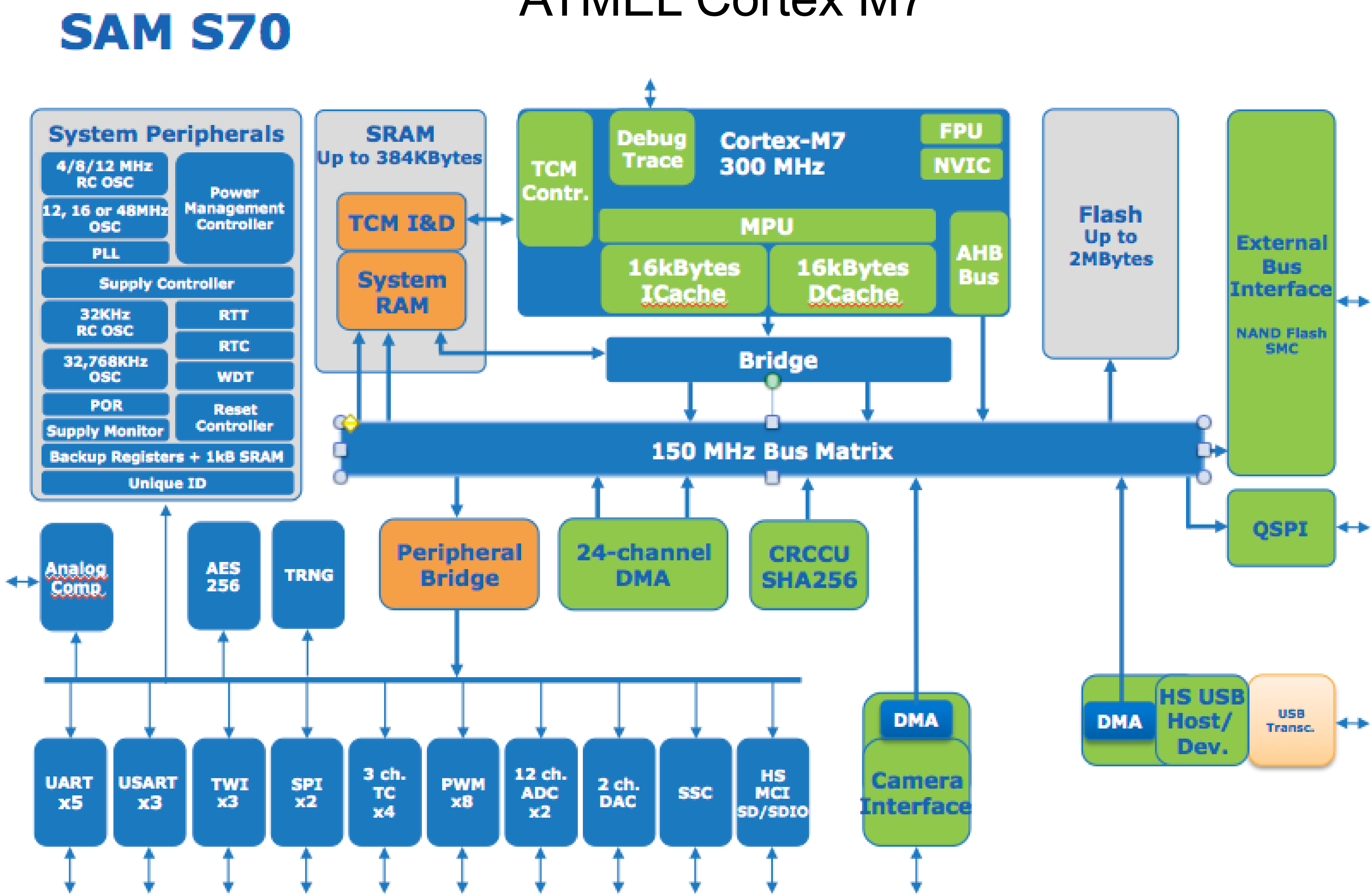
Scalable and Compatible Architecture
ARM CORTEX
Processor Technology
Cortex-M0
90 µm
Lowest cost
Low area
ARM CORTEX
Processor Technology
Cortex-M0+
15 years
Lowest power
Outstanding energy efficiency
ARM CORTEX
Processor Technology
Cortex-M3
n+1
Performance efficiency
Feature rich connectivity
ARM CORTEX
Processor Technology
Cortex-M4
Digital Signal Control (DSC)
Processor with DSP
Accelerated SIMD
Floating point (FP)
ARM CORTEX
Processor Technology
Cortex-M7
5
CoreMark per MHz
Maximum DSC Performance
Flexible Memory System
Cache, TCM, AXI, ECC
Double & Single Precision FP
Digital Signal Control application space
'8/16-bit' Traditional application space
'16/32-bit' Traditional application space


ARM® Cortex®-M7
Nested Vectored Interrupt Controller
Debug
WIC
FPU
ETM
CPU ARMv7-M
ECC
MPU
I Cache
D Cache
Data TCM
Instr TCM
AXI-M
AHB-P
AHB-S

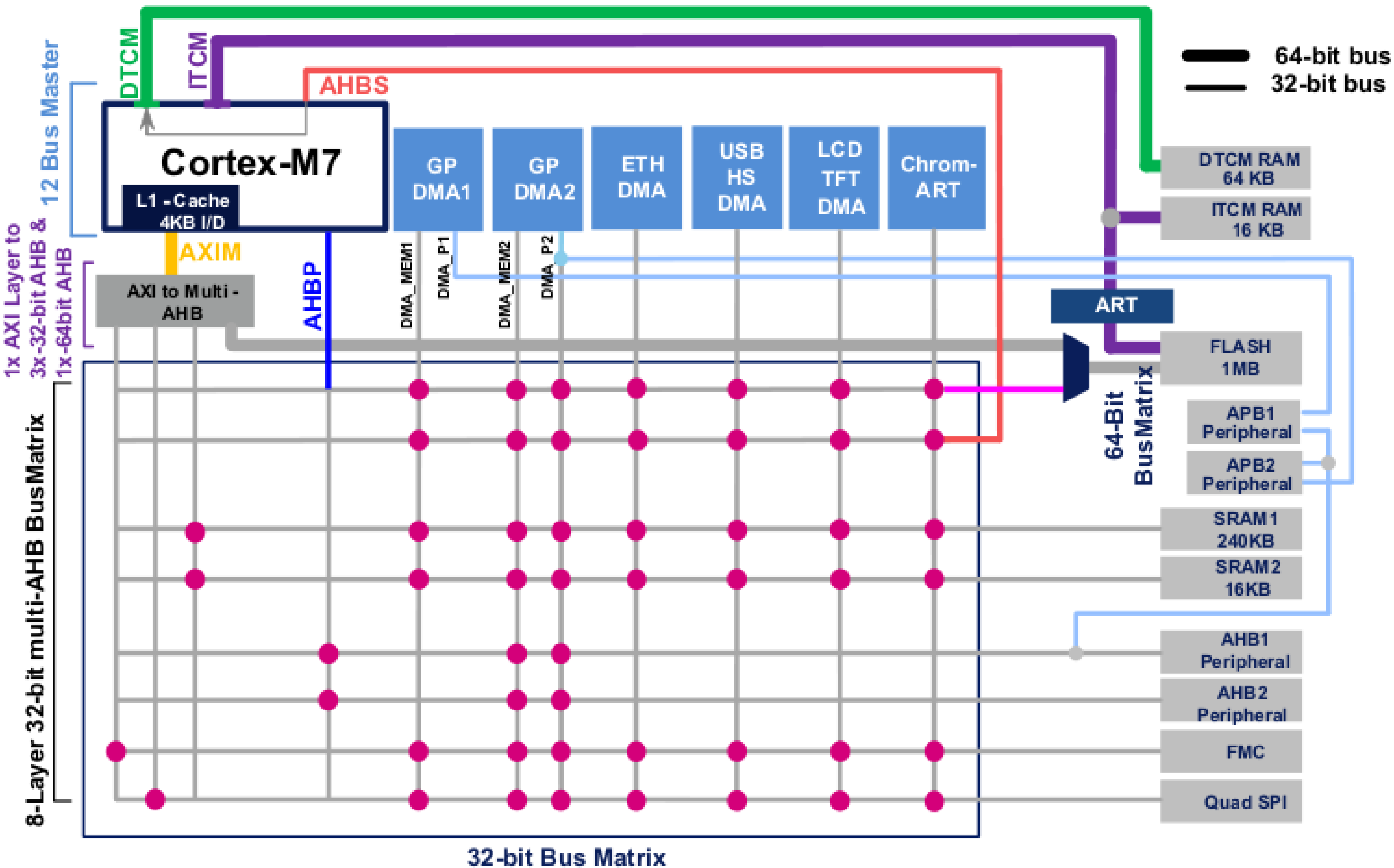
ARM Cortex-M7 Features	
ISA Support	ARMv7-M
DSP Extensions	Single cycle 16/32-bit MAC Single cycle dual 16-bit MAC 8/16-bit SIMD arithmetic Hardware Divide (2-12 Cycles)
Floating Point Unit	Single and double precision floating point unit IEEE 754 compliant
Pipeline	6-stage superscalar + branch prediction
Performance Efficiency	5 CoreMark/MHz*
Performance Efficiency	2.14 / 2.55 / 3.23 DMIPS/MHz**
Interconnect	64-bit AMBA4 AXI, AHB peripheral port (64MB to 512MB)
Instruction cache	0 to 64kB, 2-way associative with optional ECC
Data cache	0 to 64kB, 4-way associative with optional ECC
Instruction TCM	0 to 16MB with optional ECC
Data TCM	0 to 16MB with optional ECC
Memory Protection	Optional 8 or 16 region MPU with sub regions and background region
Interrupts	Non-maskable Interrupt (NMI) + 1 to 240 physical interrupts
Interrupt Priority Levels	8 to 256 priority levels
Wake-up Interrupt Controller	Up to 240 Wake-up Interrupts
Sleep Modes	Integrated WFI and WFE Instructions and Sleep On Exit capability. Sleep & Deep Sleep Signals. Optional Retention Mode with ARM Power Management Kit
Bit Manipulation	Integrated Instructions
Debug	Optional JTAG & Serial-Wire Debug Ports. Up to 8 Breakpoints and 4 Watchpoints.
Trace	Optional Instruction and Data Trace (ETM), Data Trace (DWT), and Instrumentation Trace (ITM)


System
Power supply
1.2 V regulator
POR/PDR/PVD
Xtal oscillators
32 kHz + 4 ~26 MHz
Internal RC oscillators
32 kHz + 16 MHz
PLL
Clock control
RTC/AWU
1x SysTick timer
2x watchdogs
(independent and window)
82/114/140/168 I/Os
Cyclic redundancy
check (CRC)
Control
2x 16-bit
motor control PWM
Synchronized AC timer
10x 16-bit timers
2x 32-bit timers
LP timer
Crypto/Hash processor
3DES, AES 256, GCM, CCM
SHA-1, SHA-256, MD5,
HMAC
Chrom-ART Accelerator™
ART Accelerator™
Cache I/D 4+4 Kbytes
ARM
Cortex-M7
216 MHz
Floating point unit (FPU)
Nested vector
interrupt
controller (NVIC)
JTAG/SW debug/ETM
Memory Protection Unit
(MPU)
AXI and Multi-AHB
bus matrix
16-channel DMA
True random number
generator (RNG)
1-Mbyte single
bank Flash
320-Kbyte SRAM +
16-Kbyte ITCM RAM
FMC/SRAM/NOR/NAND/
SDRAM
Dual QuadSPI
128-byte + 4-Kbyte
backup SRAM
1024-byte OTP
Connectivity
TFT LCD controller
HDMI-CEC
Camera interface
6x SPI, 3x I²S, 4x I²C
Ethernet MAC 10/100
with IEEE 1588
2x CAN 2.0B
1x USB 2.0 OTG FS/HS
1x USB 2.0 OTG FS
1x SDMMC
4x USART + 4 UART
LIN, smartcard, IrDA,
modem control
2x SAI
(Serial audio interface)
SPDIF input x4
Analog
2x 12-bit, 2-channel DACs
3x 12-bit ADC
24 channels / 2.4 MSPS
Temperature sensor

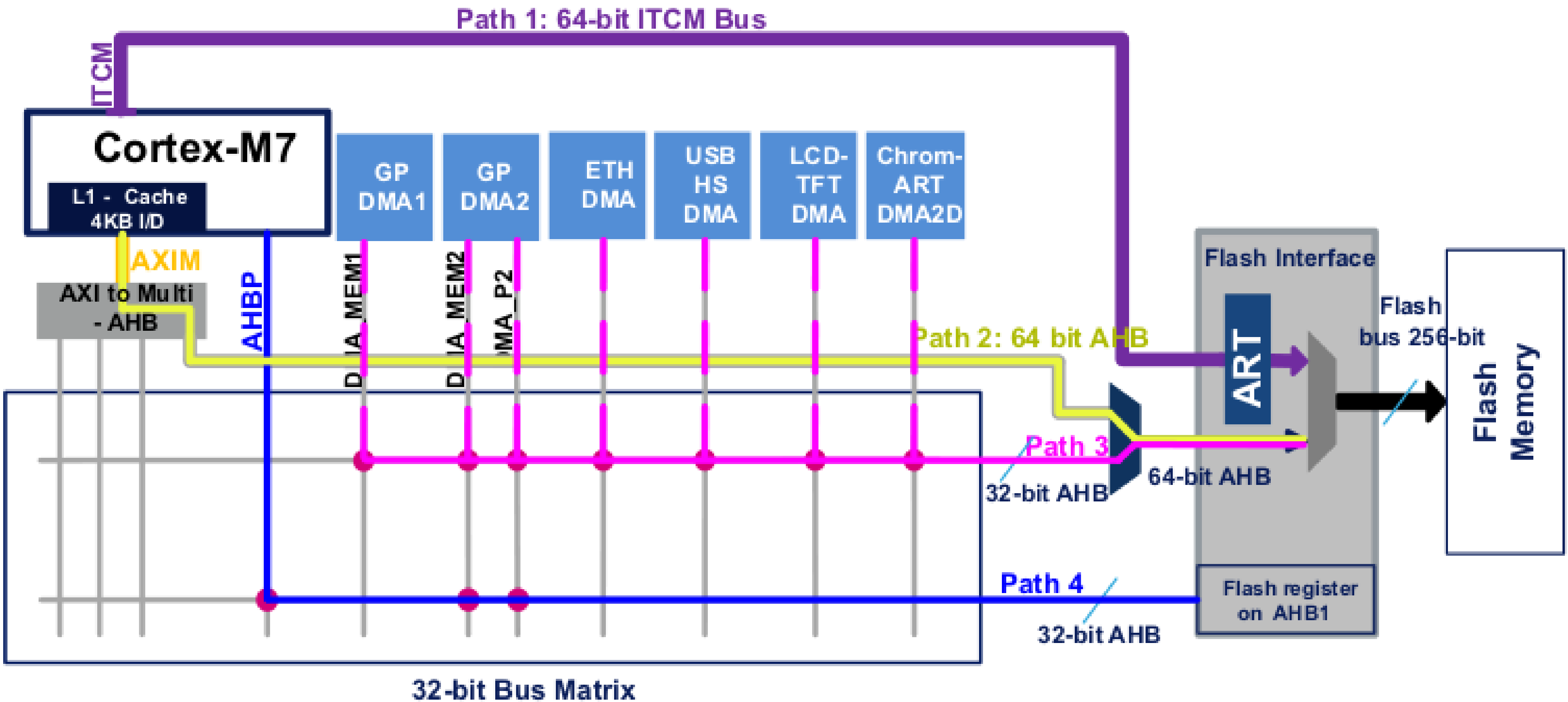
ATMEL Cortex M7



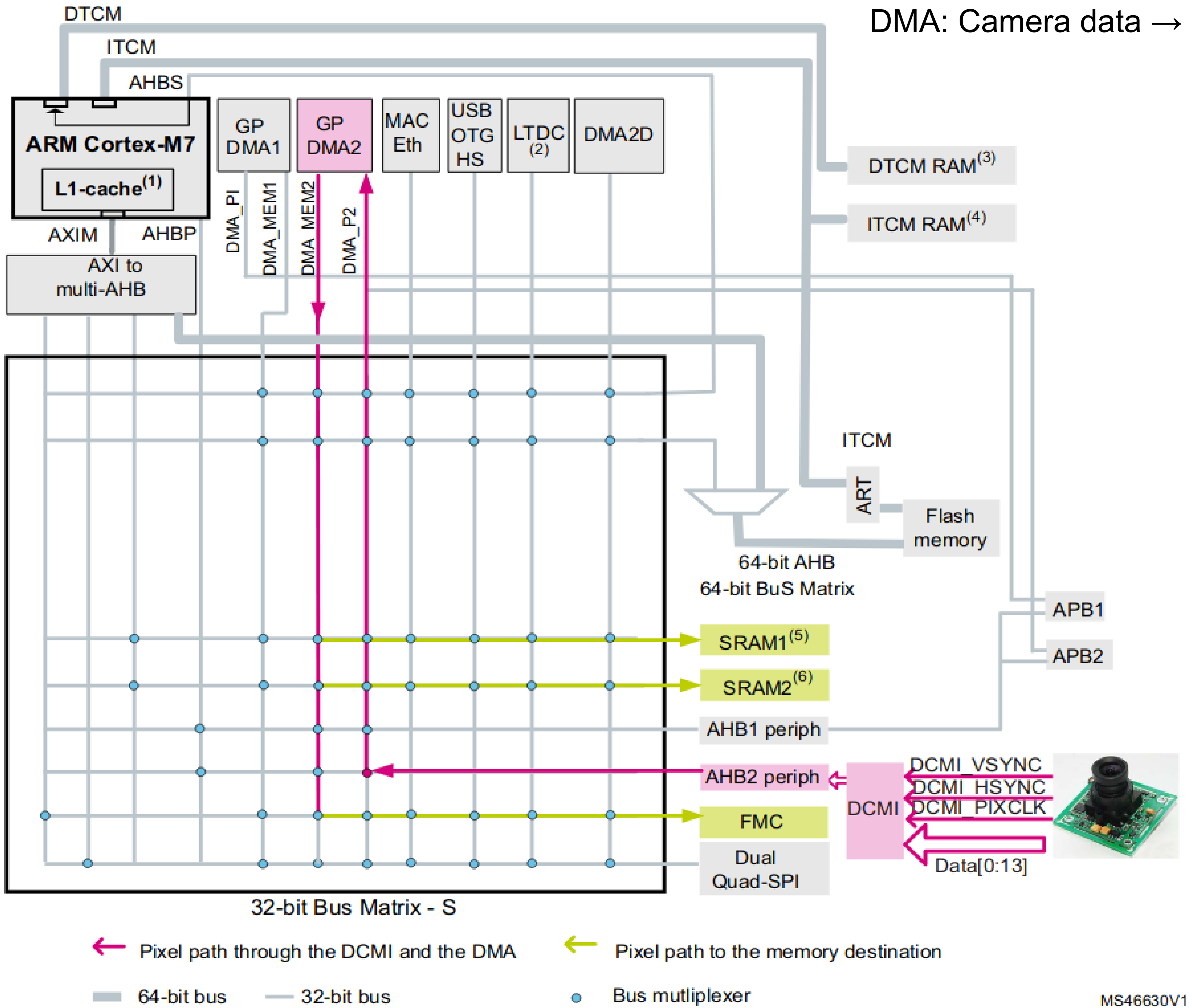
STM32F7 CM7 Bus Systems

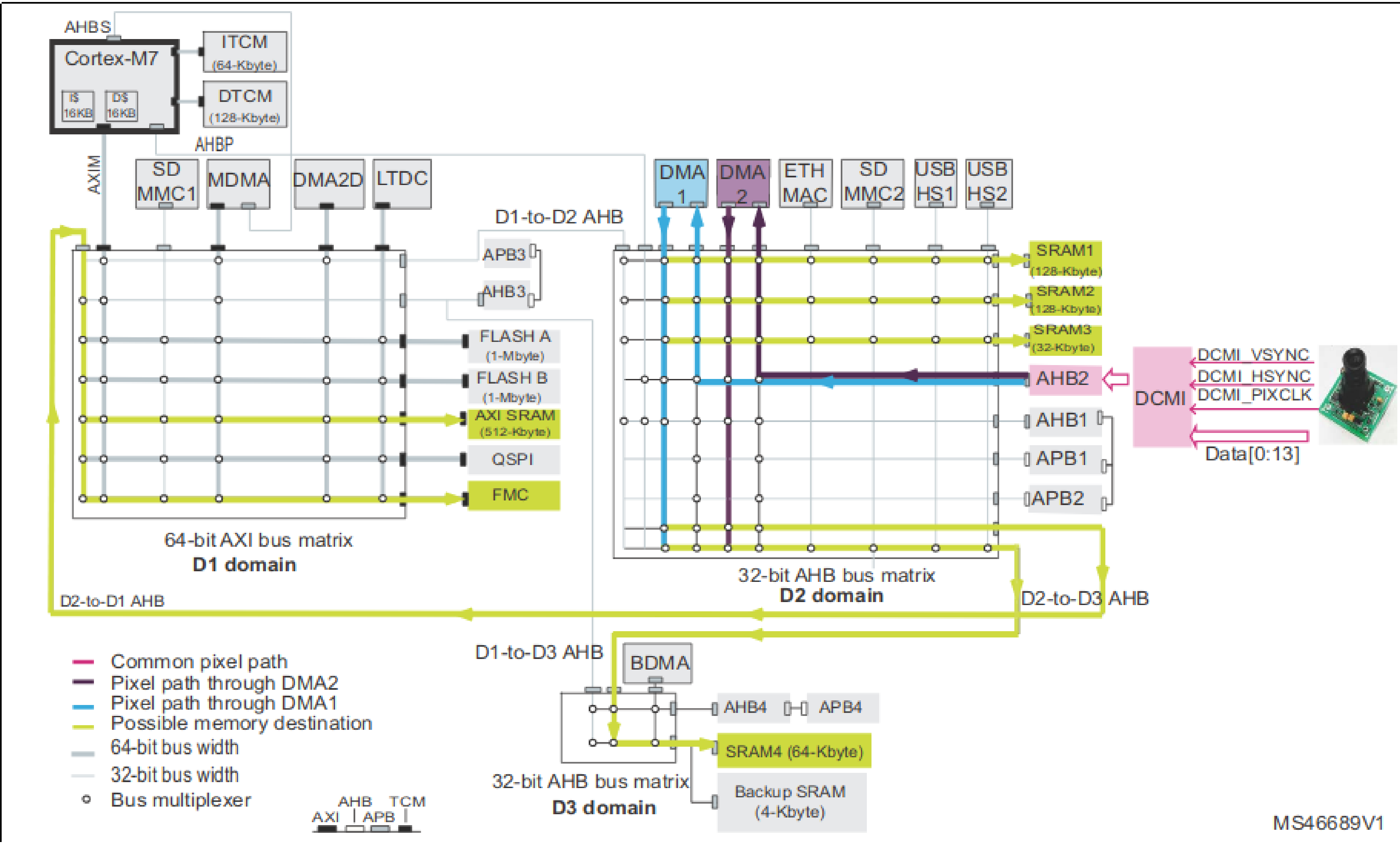


4 Paths for FLASH Access



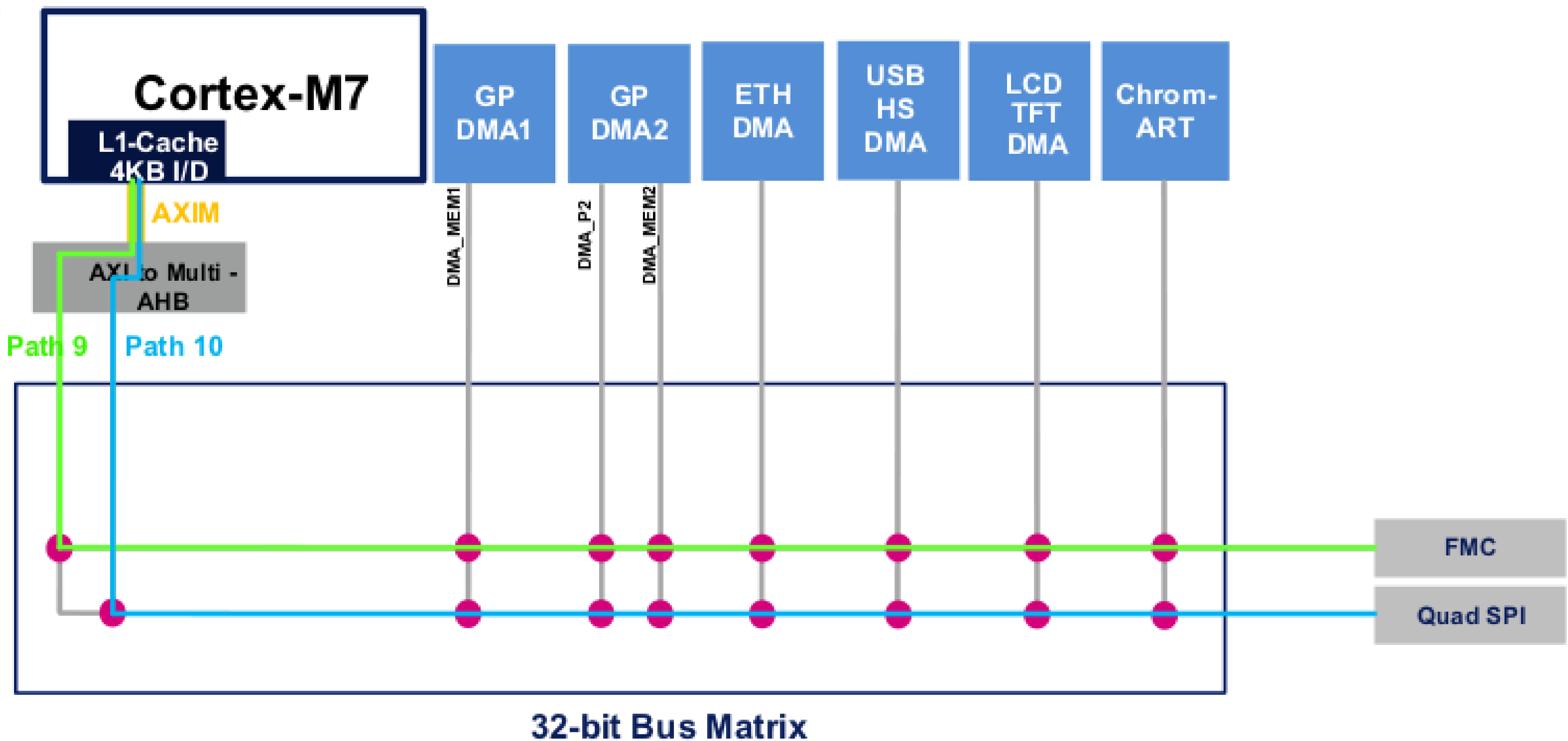
DMA: Camera data → RAM



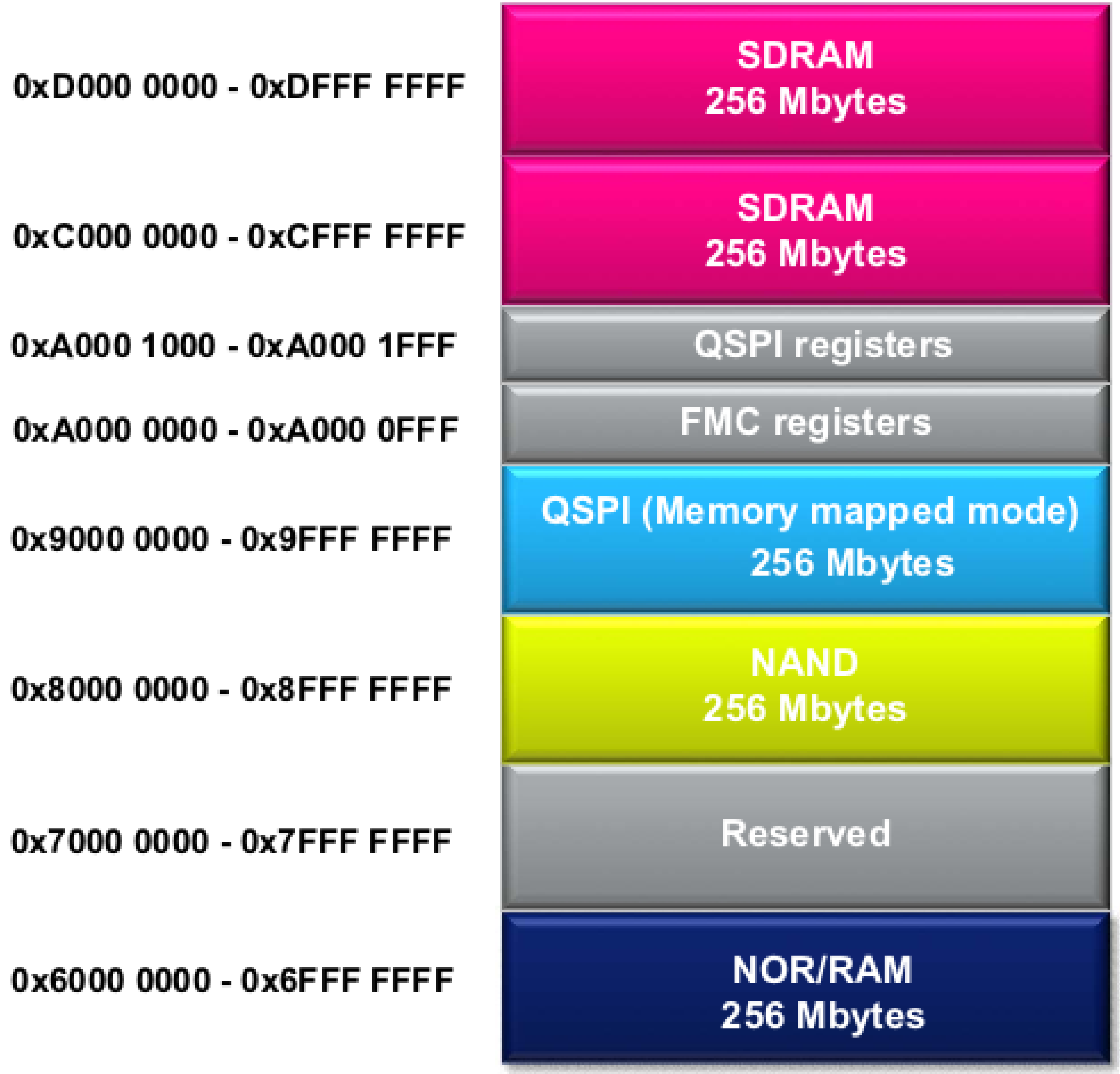


AHBS
Cortex-M7
I$ 16KB
D$ 16KB
ITCM (64-Kbyte)
DTCM (128-Kbyte)
AHBP
AXIM
SD MMC1
MDMA
DMA2D
LTDC
D1-to-D2 AHB
DMA 1
DMA 2
ETH MAC
SD MMC2
USB HS1
USB HS2
APB3
AHB3
FLASH A (1-Mbyte)
FLASH B (1-Mbyte)
AXI SRAM (512-Kbyte)
QSPI
FMC
SRAM1 (128-Kbyte)
SRAM2 (128-Kbyte)
SRAM3 (32-Kbyte)
AHB2
AHB1
APB1
APB2
DCMI
DCMI_VSYNC
DCMI_HSYNC
DCMI_PIXCLK
Data[0:13]
64-bit AXI bus matrix
D1 domain
32-bit AHB bus matrix
D2 domain
D2-to-D1 AHB
D2-to-D3 AHB
D1-to-D3 AHB
BDMA
AHB4
APB4
SRAM4 (64-Kbyte)
Backup SRAM (4-Kbyte)
32-bit AHB bus matrix
D3 domain
Common pixel path
Pixel path through DMA2
Pixel path through DMA1
Possible memory destination
64-bit bus width
32-bit bus width
Bus multiplexer
AXI
AHB
APB
TCM
MS46689V1

External Memory Interfaces

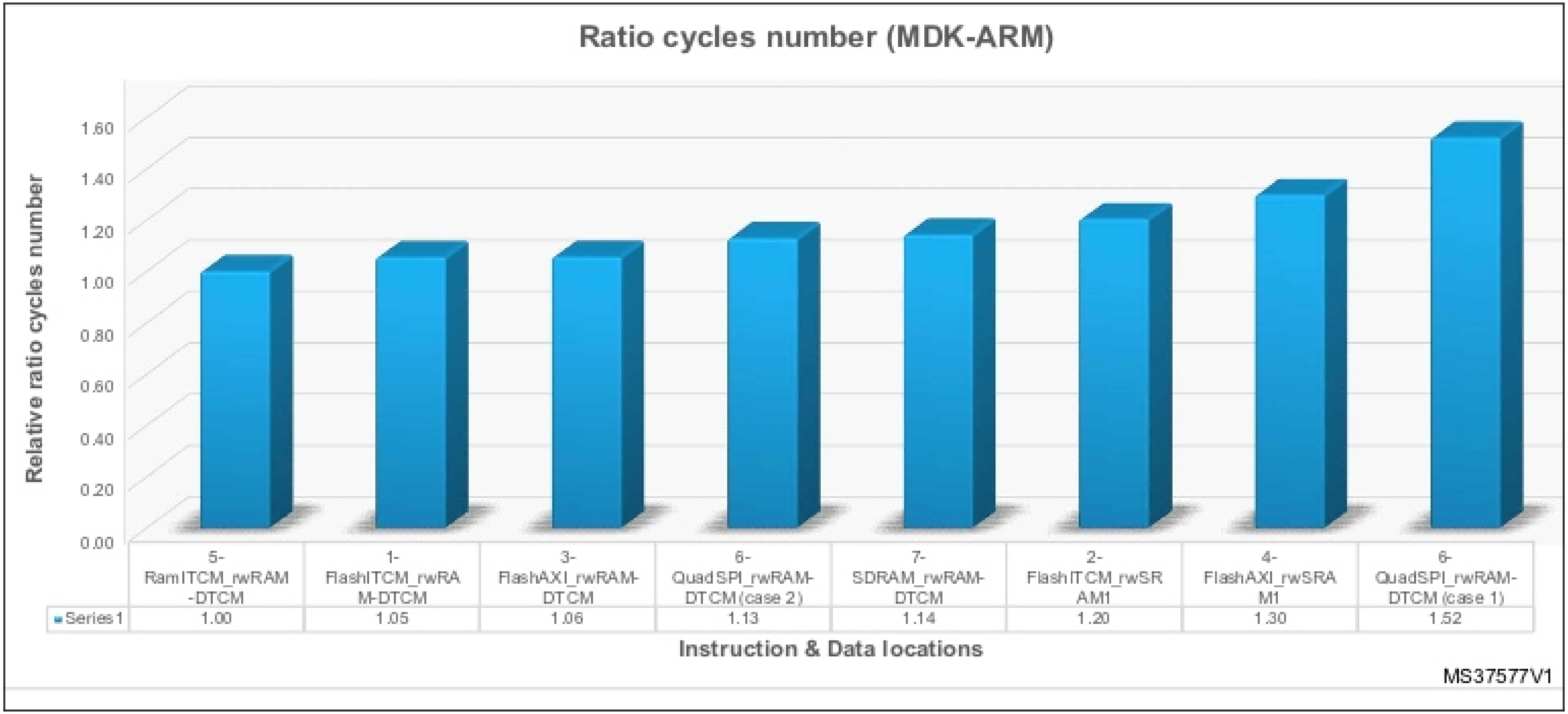


External memory mapping



FFT Execution Time

Figure 9. FFT relative ratio cycle number with MDK-ARM



www.st.com/st-web-ui/static/active/en/resource/technical/document/application_note/DM00169764.pdf

Comparing CM4 and CM7

step = 1 / 2   diff = -130361 ppm   runtime = 18 micro-seconds
step = 1 / 4   diff = -46436 ppm   runtime = 40 micro-seconds
step = 1 / 8   diff = -16480 ppm   runtime = 85 micro-seconds
step = 1 / 16   diff = -5837 ppm   runtime = 172 micro-seconds
step = 1 / 32   diff = -2065 ppm   runtime = 348 micro-seconds
step = 1 / 64   diff = -730 ppm   runtime = 699 micro-seconds
step = 1 / 128   diff = -258 ppm   runtime = 1399 micro-seconds
step = 1 / 256   diff = -91 ppm   runtime = 2796 micro-seconds
step = 1 / 512   diff = -32 ppm   runtime = 5591 micro-seconds
step = 1 / 1024   diff = -11 ppm   runtime = 11161 micro-seconds
step = 1 / 2048   diff = -4 ppm   runtime = 22319 micro-seconds
step = 1 / 4096   diff = -1 ppm   runtime = 44652 micro-seconds
step = 1 / 8192   diff = 0 ppm   runtime = 89282 micro-seconds
step = 1 / 16384   diff = 0 ppm   runtime = 178575 micro-seconds
step = 1 / 32768   diff = 0 ppm   runtime = 357095 micro-seconds
step = 1 / 65536   diff = 0 ppm   runtime = 714186 micro-seconds
step = 1 / 131072   diff = 0 ppm   runtime = 1428129 micro-seconds
step = 1 / 262144   diff = 0 ppm   runtime = 2855990 micro-seconds
step = 1 / 524288   diff = 0 ppm   runtime = 5711562 micro-seconds
step = 1 / 1048576   diff = 0 ppm   runtime = 11422791 micro-seconds
step = 1 / 2097152   diff = 0 ppm   runtime = 22844415 micro-seconds
step = 1 / 4194304   diff = 0 ppm   runtime = 45687856 micro-seconds
step = 1 / 8388608   diff = 0 ppm   runtime = 91372910 micro-seconds
step = 1 / 16777216   diff = 0 ppm   runtime = 182743030 micro-seconds
step = 1 / 33554432   diff = 0 ppm   runtime = 365485955 micro-seconds

step = 1 / 2   diff = -130361 ppm   runtime = 3 micro-seconds
step = 1 / 4   diff = -46436 ppm   runtime = 2 micro-seconds
step = 1 / 8   diff = -16480 ppm   runtime = 4 micro-seconds
step = 1 / 16   diff = -5837 ppm   runtime = 7 micro-seconds
step = 1 / 32   diff = -2065 ppm   runtime = 13 micro-seconds
step = 1 / 64   diff = -730 ppm   runtime = 24 micro-seconds
step = 1 / 128   diff = -258 ppm   runtime = 47 micro-seconds
step = 1 / 256   diff = -91 ppm   runtime = 94 micro-seconds
step = 1 / 512   diff = -32 ppm   runtime = 186 micro-seconds
step = 1 / 1024   diff = -11 ppm   runtime = 370 micro-seconds
step = 1 / 2048   diff = -4 ppm   runtime = 747 micro-seconds
step = 1 / 4096   diff = -1 ppm   runtime = 1478 micro-seconds
step = 1 / 8192   diff = 0 ppm   runtime = 2953 micro-seconds
step = 1 / 16384   diff = 0 ppm   runtime = 5904 micro-seconds
step = 1 / 32768   diff = 0 ppm   runtime = 11810 micro-seconds
step = 1 / 65536   diff = 0 ppm   runtime = 23619 micro-seconds
step = 1 / 131072   diff = 0 ppm   runtime = 47234 micro-seconds
step = 1 / 262144   diff = 0 ppm   runtime = 94469 micro-seconds
step = 1 / 524288   diff = 0 ppm   runtime = 188941 micro-seconds
step = 1 / 1048576   diff = 0 ppm   runtime = 377881 micro-seconds
step = 1 / 2097152   diff = 0 ppm   runtime = 755758 micro-seconds
step = 1 / 4194304   diff = 0 ppm   runtime = 1511517 micro-seconds
step = 1 / 8388608   diff = 0 ppm   runtime = 3023028 micro-seconds
step = 1 / 16777216   diff = 0 ppm   runtime = 6046052 micro-seconds
step = 1 / 33554432   diff = 0 ppm   runtime = 12092096 micro-seconds

→ CM7 is 30 times faster than the CM4 with it's double-precision SW emulation !

STM32F7 Cube MX Software